Design and simulation of Integrated Photonic Interferometers for experimental extraction of waveguide group index

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***Abstract*—This work presents the design and simulation of integrated photonic interferometers for experimental extraction of the group index of waveguides on SOI platform. The primary objective is to develop interferometric test structures whose free spectral range (FSR) can be used to determine the wavelength-dependent group index and subsequently compare the experimentally extracted values with numerical simulations. A silicon-on-insulator (SOI) platform with a 220-nm-thick silicon device layer is considered. The baseline waveguide is a 500-nm-wide strip waveguide, and both quasi-transverse-electric (TE) and quasi-transverse-magnetic (TM) polarizations are investigated. Nine photonic circuits are proposed, consisting of Mach–Zehnder interferometers (MZIs) as well as Michelson interferometers. Variations in interferometer path-length imbalance are included to investigate their influence on FSR and group index. The wavelength-dependent effective index obtained from Lumerical MODE Solutions is converted into a polynomial waveguide compact model and implemented in Lumerical INTERCONNECT. The compact model will first be validated independently using an Optical Network Analyzer (ONA), from which group delay and hence group index is extracted. The FSR obtained from the simulated transmission spectra will be used to independently extract the group index.**

***Index Terms*— group index, integrated photonics, Mach–Zehnder interferometer, Michelson interferometer, optical network analyser.**

I. INTRODUCTION

Integrated photonic circuits rely on accurate knowledge of waveguide optical properties for the design and analysis of passive photonic components. Among these properties, the effective index and group index are particularly important because they determine optical propagation delay, dispersion, and the spectral characteristics of interferometric circuits.

The objective of this project is to design integrated photonic interferometers that can be used as experimental structures for extracting the group index of SOI strip waveguides. The group index obtained from the interferometer free spectral range will be compared with values obtained independently from numerical waveguide simulations.

The proposed structures are based primarily on Mach–Zehnder interferometers (MZIs), with Michelson interferometers included as an additional interferometric architecture.

A major objective of the simulation methodology is to establish a complete validation chain:

MODE waveguide simulation → Dispersion Analysis →
waveguide compact model →
ONA validation →
ONA validation of Ebeam waveguide
Interferometer simulation → FSR extraction → group-index extraction→
Comparison of extracted n_g

II. DESIGN OBJECTIVES

The primary objectives of the proposed work are:

1) To design silicon waveguides on a 220-nm SOI platform.
2) To obtain the wavelength-dependent effective index using Lumerical MODE Solutions.
3) To calculate the waveguide group index from the simulated effective-index dispersion.
4) To develop a polynomial compact model of the waveguide.
5) To implement the compact model in Lumerical INTERCONNECT.
6) To validate the compact model using an Optical Network Analyzer.
7) To extract group delay and group index from the ONA Results.
8) To design MZI and Michelson interferometers with suitable path-length imbalance.
9) To simulate interferometer transmission spectra.
10) To extract group index from the simulated FSR.
11) To investigate the effect of path-length imbalance on FSR.

III. WAVEGUIDE DESIGN

$$Si - Core_{\text{height}} = 220nm, Si - Core_{\text{width}} = 500nm$$

Both quasi-TE and quasi-TM fundamental modes are considered.
The wavelength region of interest is centred around $\lambda_0 = 1550$ nm.

IV. WAVEGUIDE MODE SIMULATION,DISPERSION ANALYSIS & WAVEGUIDE COMPACT MODEL

The waveguide modes are simulated using Lumerical MODE Solutions.

$$Si - Core_{\text{height}} = 220nm, Si - Core_{\text{width}} = 500nm$$
$$mesh\ step = 10nm$$
$$Boundary\ Conditions\ of\ FDE\ Mode\ Solver = Metal$$

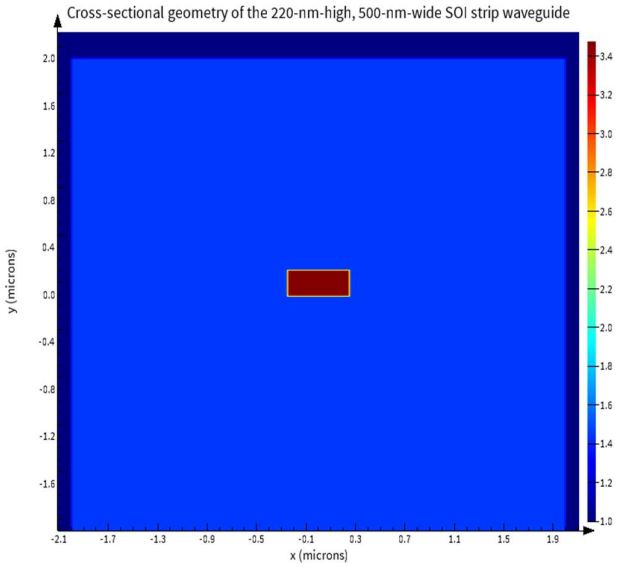


Fig. 1. Cross-sectional geometry of the 220-nm-high, 500-nm-wide SOI strip waveguide**.**

Eigensolver Analysis

Mode list

mode #	effective index	wavelength (μm)	loss (dB/cm)	group index	TE polarization fraction (Ex)
1	2.444348+1.247409e-09i	1.55	0.00043921	4.196081+2.663326e-09i	98
2	1.771592+8.530316e-10i	1.55	0.00030035	3.750631+4.885962e-09i	4

Fig. 2. EigenSolver Analysis Window depicting fundamental TE and TM modal parameters

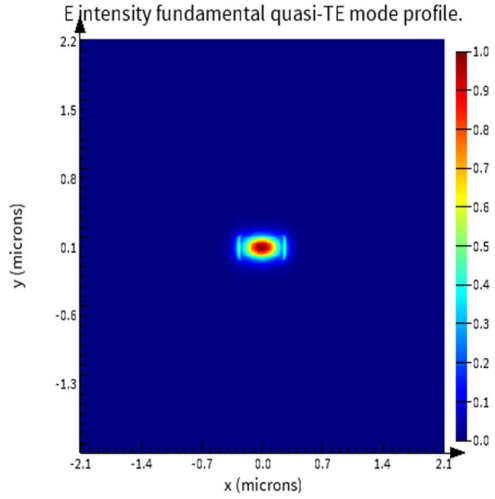


Fig. 3. Simulated fundamental quasi-TE mode profile.

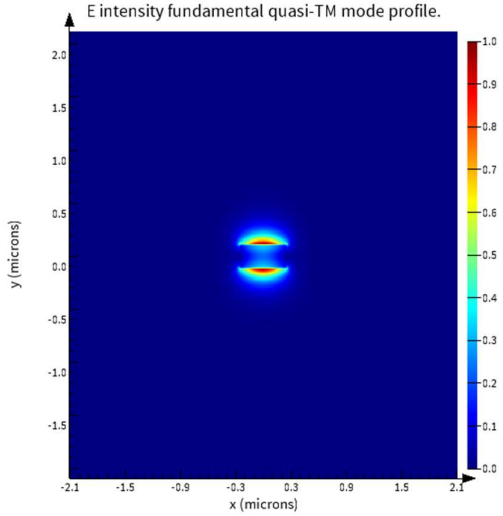
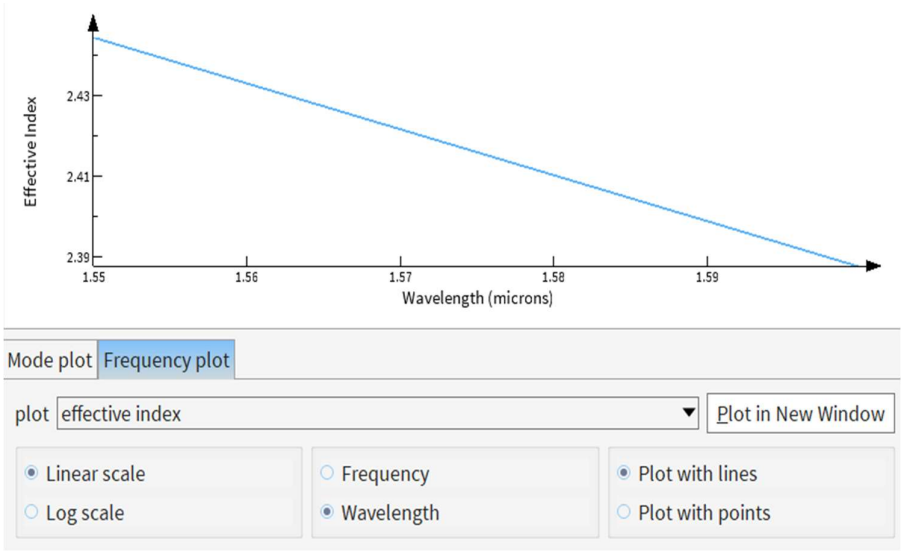


Fig. 4. Simulated fundamental quasi-TM mode profile.



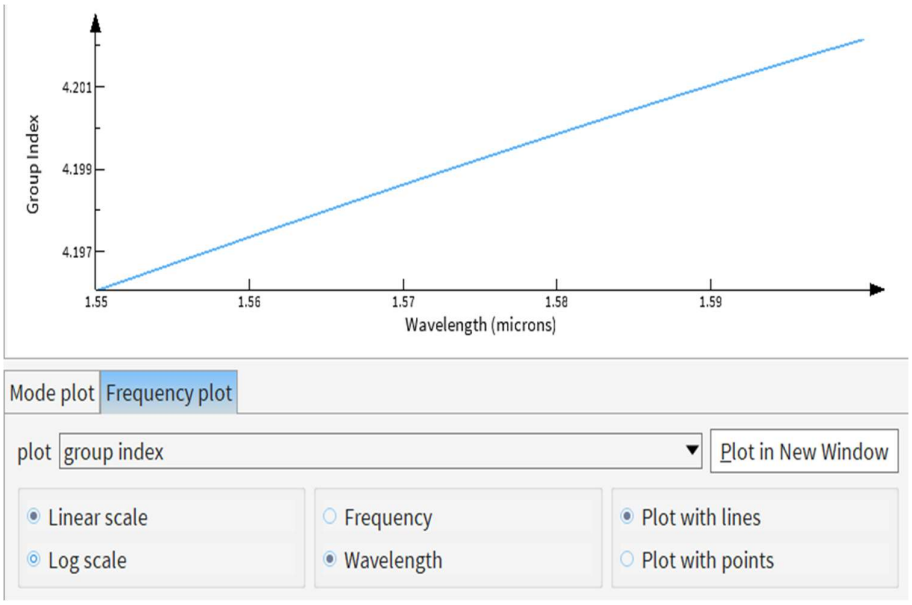
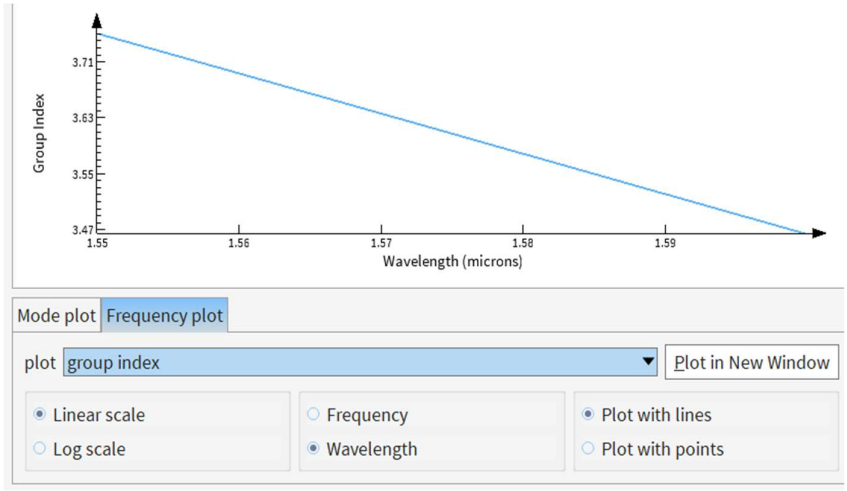


Fig. 5. Dispersion Analysis of fundamental quasi-TE mode.



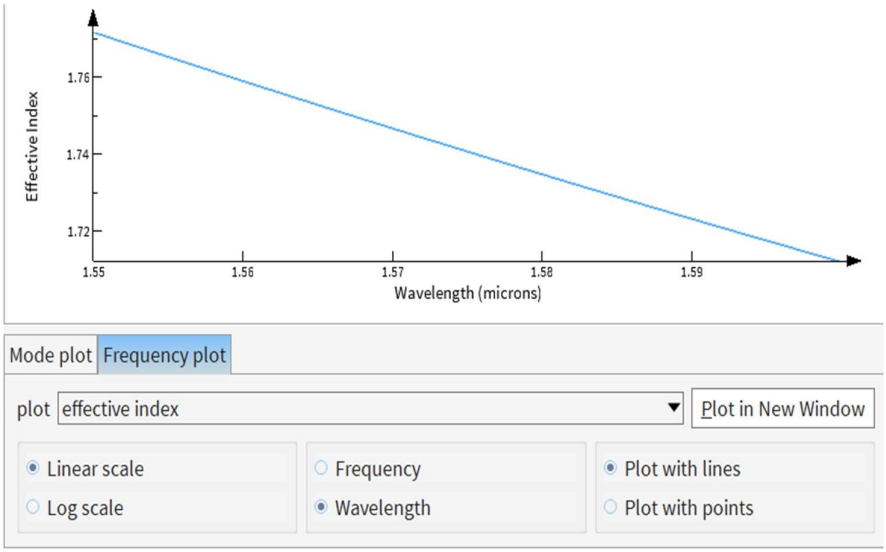


Fig. 6. Dispersion Analysis of fundamental quasi-TM mode.

For circuit-level simulation, the simulated effective index will be represented using a polynomial compact model and is mentioned here.

$$n_{\text{eff-TE-MODE}}(\lambda) = 2.44435 - 1.13022(\lambda - \lambda_0) - 0.0387076(\lambda - \lambda_0)^2$$

$$\boldsymbol{n_{eff-TM-MODE}(\lambda) = 1.77159 - 1.27651(\lambda - \lambda_0) + 1.81631(\lambda - \lambda_0)^2}$$

$$\boldsymbol{n_g(\lambda) = \; n_{eff}(\lambda) - \lambda \frac{dn_{eff}(\lambda)}{d\lambda}}$$

$$\boldsymbol{n_{g-TE-MODE}(\lambda) = 2.44435 + 1.13022(\lambda_0) + 0.0387076\left(\lambda^2 - {\lambda_0}^2\right)}$$

$$\boldsymbol{n_{g-TM-MODE}(\lambda) = 1.77159 + 1.27651(\lambda_0) - 1.81631\left(\lambda^2 - {\lambda_0}^2\right)}$$

Where λ is in microns

V. COMPACT MODEL VALIDATION USING LUMERICAL INTERCONNECT ONA

A. ONA Configuration

Before connecting the waveguide compact model to an interferometer, it is independently validated using the Optical Network Analyzer in Lumerical INTERCONNECT as shown below for Simulated model and Ebeam model separately (for both TE and TM polarization).

The ONA will sweep wavelength over the required measurement range.

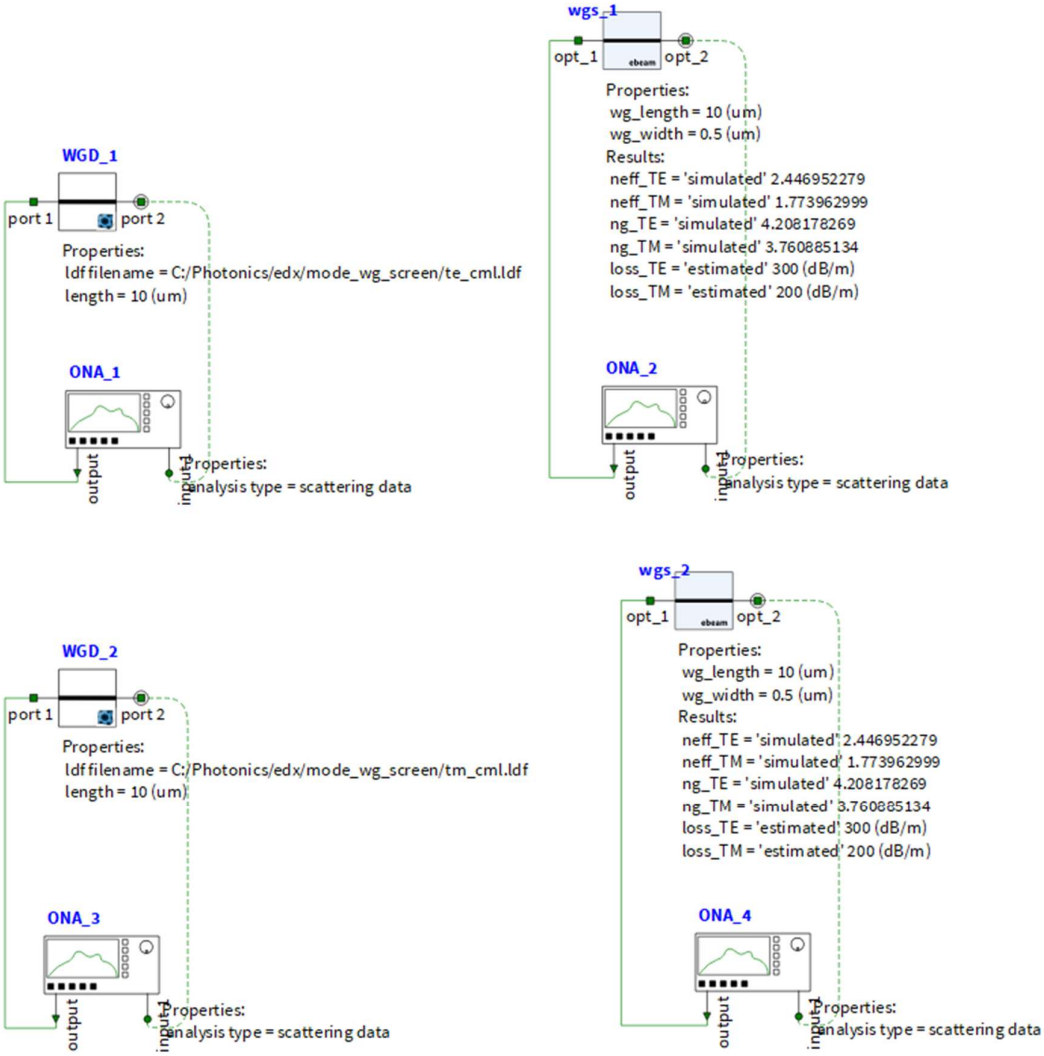


Fig. 7. Lumerical IC setup for CML validation

B. Group Delay From ONA

The group delay can be extracted as mentioned below.

For a waveguide of physical length L,

$$\boxed{n_g = \frac{c\tau_g}{L}}$$

where $c$is the speed of light and τ_g is the group delay.

This provides an independent circuit-level extraction of the group index.

TABLE I
ONA Simulation Results Summary

$n_{\mathrm{g-TE-MODE}}$	$\tau_{\mathrm{g}-TE-MODE-IC}$	$n_{\mathrm{g-TE-MODE-IC}}$	$\tau_{\mathrm{g}-TE-Ebeam-IC}$	$n_{\mathrm{g-TE-Ebeam-IC}}$
4.1962	13.9966 ns/m	4.196075	14.0368 ns/m	4.2081
$n_{\mathrm{g-TM-MODE}}$	$\tau_{\mathrm{g}-TM-MODE-IC}$	$n_{\mathrm{g-TM-MODE-IC}}$	$\tau_{\mathrm{g}-TM-Ebeam-IC}$	$n_{\mathrm{g-TM-Ebeam-IC}}$
3.75	12.5102 ns/m	3.75	12.5136 ns/m	3.75

VI. MACH–ZEHNDER INTERFEROMETER THEORY

The MZI consists of an input splitter, two arms with different lengths, and an output combiner. The path-length difference is

$$\boldsymbol{\Delta L = L_2 - L_1}.$$

Phase difference between the two MZI arms is

$$\boldsymbol{\Delta\phi(\lambda) = \frac{2\pi n_{eff}(\lambda)\Delta L}{\lambda}}.$$

For an unbalanced MZI(assuming lossless case), the transmission can be represented as

$$\boldsymbol{I_o(\lambda) = \frac{I_i[1 + cos(\Delta\phi(\lambda))]}{2}}$$

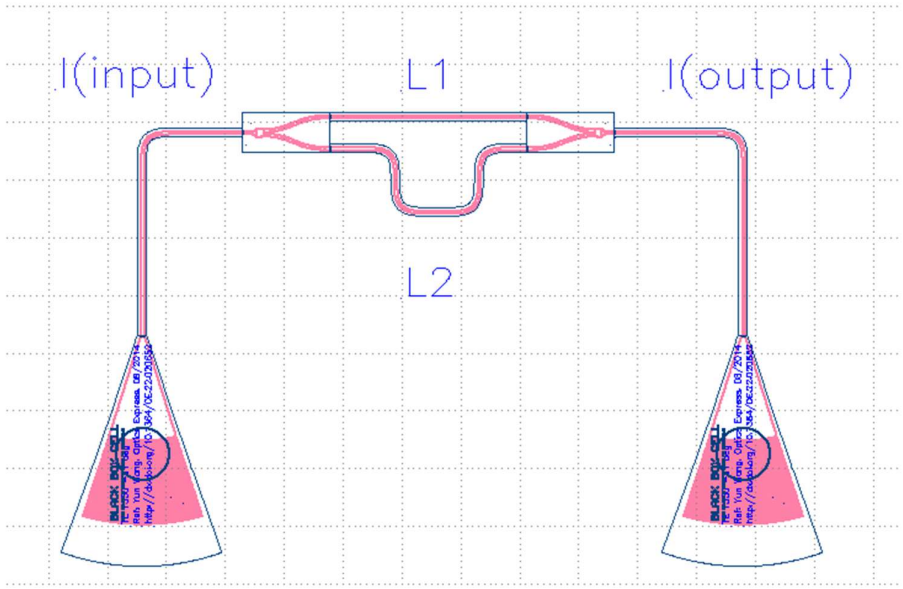


Fig. 8. MZI structure

Sample MZI Transmission looks as below.

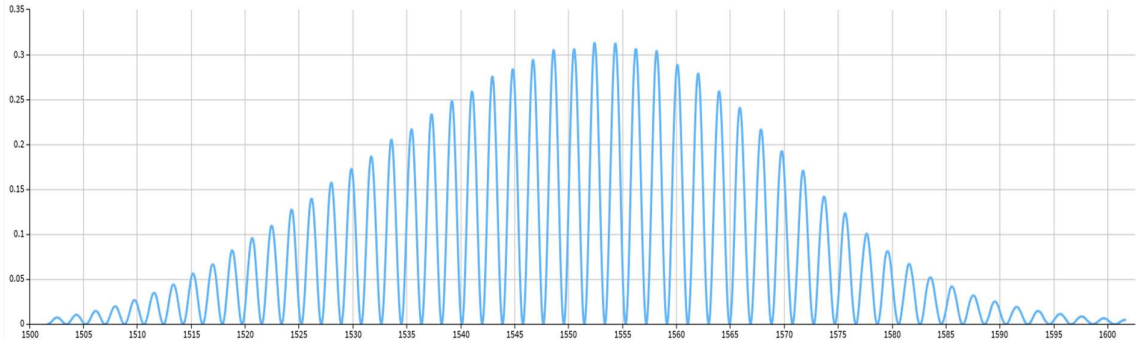


Fig. 9. Transmission Spectrum of MZI structure with Grating Coupler

VII. GROUP-INDEX EXTRACTION FROM MZI FSR

Adjacent maxima occur when the phase difference changes by 2π:

$$\boldsymbol{\Delta\phi(\lambda + FSR) - \Delta\phi(\lambda) = 2\pi}.$$

Using the definition of group index gives the standard MZI FSR relationship

$$\boxed{\boldsymbol{FSR \approx \frac{\lambda^2}{n_g \Delta L}}}$$

and therefore

$$\boxed{\boldsymbol{n_g \approx \frac{\lambda^2}{\Delta L\, FSR}}}.$$

This equation is the principal relation used to extract the waveguide group index from the MZI transmission spectrum.

VIII. DESIGN OF MZI PATH-LENGTH DIFFERENCE

The measurement bandwidth is approximately 50 nm. The path-length difference must therefore be selected so that several interference fringes are observable within the available bandwidth.

As an initial design estimate, taking

$$n_{g-TE} \approx 4.2, n_{g-TM} \approx 3.75$$

near 1550 nm gives

$$\boldsymbol{FSR_{TE} = \frac{1.55^2}{4.2\Delta L}, FSR_{TM} = \frac{1.55^2}{3.75\Delta L}}$$

TABLE II
DESIGN-CALCULATED MZI FSR FOR DIFFERENT PATH-LENGTH DIFFERENCES

ΔL	Polarization	Calculated FSR at 1550 nm
100 µm	TE	5.72
150 µm	TE	3.81
200 µm	TE	2.86
250 µm	TE	2.29
250 µm	TM	2.56

IX. INTERFEROMETER SIMULATION

The complete optical circuit includes input and output grating couplers. This represents the optical path that more closely resembles the experimental fiber-to-chip measurement.

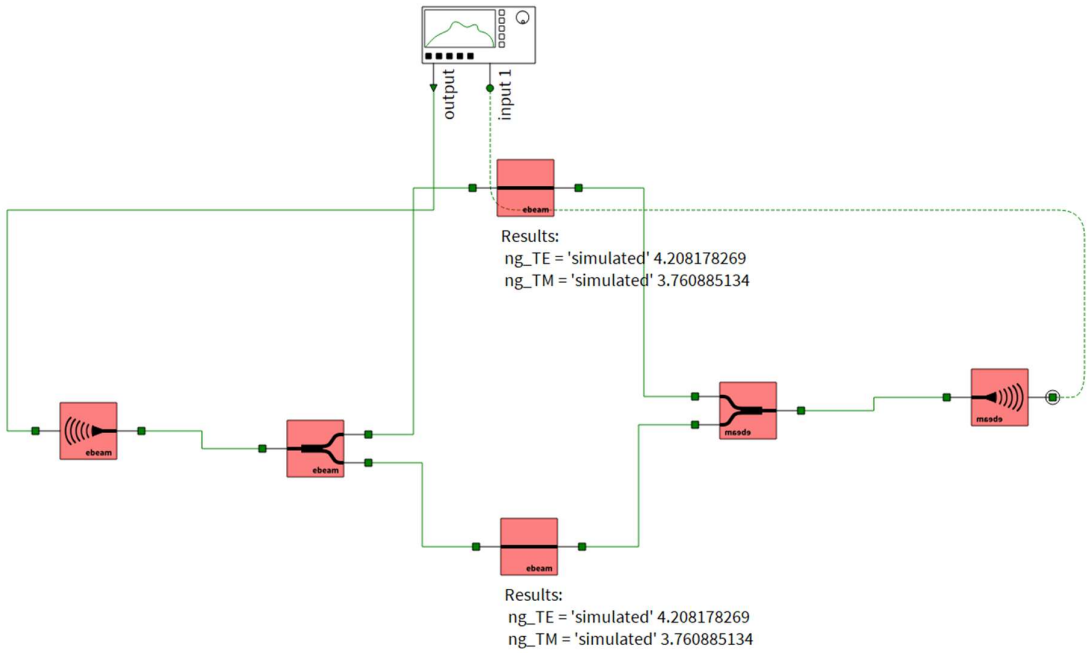


Fig. 10. Simulation of MZI structure with Grating Coupler

X. PROPOSED PHOTONIC DESIGNS

The following nine designs are selected for fabrication and characterization.

TABLE III
PROPOSED DESIGN SET

S No.	Interferometer Design	Pol	Length	Remarks
1	MZI	TE	100 µm	Same design to fabricated twice to test manufacturing variability.
2	MZI	TE	100 µm	
3	MZI	TE	150 µm	FSR vs ΔL
4	MZI	TE	200 µm	
5	MZI	TE	250 µm	
6	TE Cutback	TE	--	
7	MZI	TM	250 µm	Impact of fabrication on relatively weakly guiding mode wrt TE
8	MICHELSON	TE	345.603 µm	
9		TM	748.879 µm	
10	TM Cutback	TM		

XI. MICHELSON INTERFEROMETER

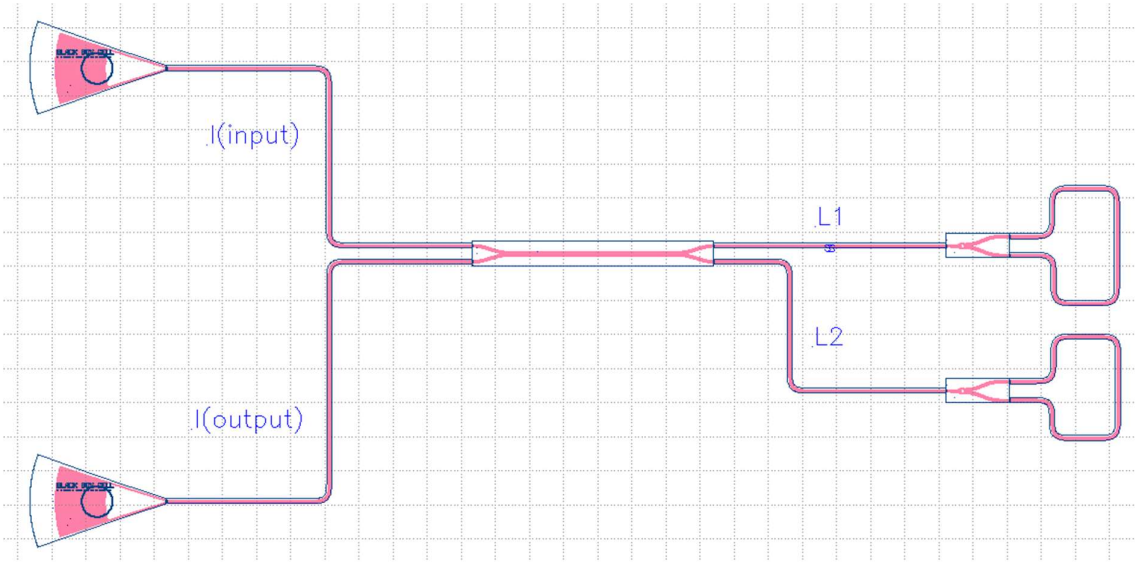


Fig. 11. Michelson Interferometer structure

For the Michelson geometry, the optical field travels through the arm in both directions. Therefore, the phase difference is

$$\Delta\phi(\lambda) = \frac{4\pi n_{eff}(\lambda)\Delta L}{\lambda}.$$

The corresponding FSR is

$$\boxed{FSR \approx \frac{\lambda^2}{2n_g\Delta L}}$$

and therefore

$$\boxed{n_g \approx \frac{\lambda^2}{2\Delta L\, FSR}}.$$

The factor of two is important and is explicitly considered during experimental analysis.

XII. MICHELSON DESIGN PARAMETERS

TABLE IV

ΔL	Polarization	Calculated FSR at 1550 nm
345.603 µm	TE	0.828
748.879 µm	TM	0.428

XIII. INTERCONNECT SIMULATION SETUP

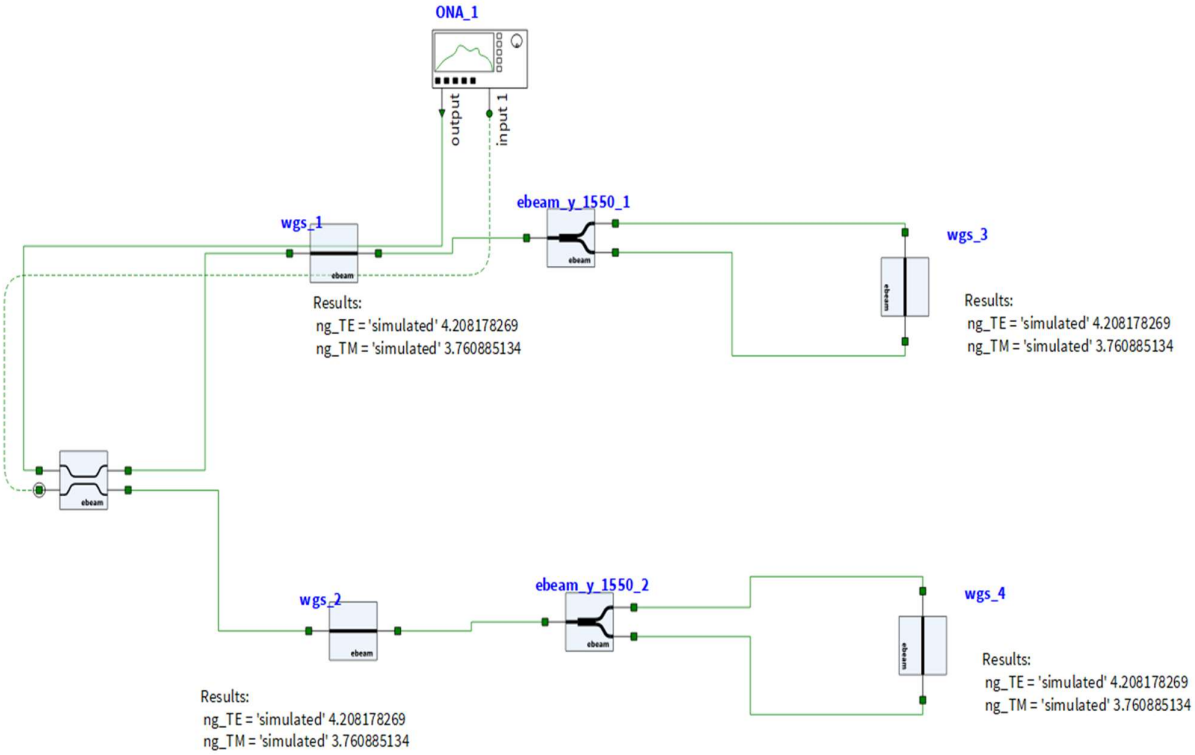


Fig. 12. Simulation of Michelson Interferometer structure

TABLE V
SUMMARY OF SIMULATION RESULTS

S No 1 & 2 : *wgs_1 length: 60.000 um* *wgs_2 length: 160.000 um* *Delta L = abs(L2 - L1): 100.000 um* *Peak transmission: 1548.601263 nm, 1554.342130 nm* *Transmission FSR: 5.740867 nm at 1551.471697 nm* *ng from peak transmission: 4.192859* *ONA FSR sample nearest 1550 nm: 1548.601263 nm* *ONA FSR: 5.678798 nm* *ng from ONA FSR: 4.223017*	***S No 3 :*** *wgs_1 length: 60.000 um* *wgs_2 length: 210.000 um* *Delta L = abs(L2 - L1): 150.000 um* *Peak transmission: 1548.591281 nm, 1552.363599 nm* *Transmission FSR: 3.772317 nm at 1550.477440 nm* *ng from peak transmission: 4.248459* *ONA FSR sample nearest 1550 nm: 1548.591281 nm* *ONA FSR: 3.773895 nm* *ng from ONA FSR: 4.236357*
S No 4 *:* *wgs_1 length: 60.000 um* *wgs_2 length: 260.000 um* *Delta L = abs(L2 - L1): 200.000 um* *Peak transmission: 1548.581300 nm, 1551.441339 nm* *Transmission FSR: 2.860039 nm at 1550.011319 nm* *ng from peak transmission: 4.200178* *ONA FSR sample nearest 1550 nm: 1548.581300 nm* *ONA FSR: 2.849514 nm* *ng from ONA FSR: 4.207918*	***S No 5 :*** *wgs_1 length: 60.000 um* *wgs_2 length: 310.000 um* *Delta L = abs(L2 - L1): 250.000 um* *Peak transmission: 1548.581300 nm, 1550.860477 nm* *Transmission FSR: 2.279178 nm at 1549.720889 nm* *ng from peak transmission: 4.214915* *ONA FSR sample nearest 1550 nm: 1550.860477 nm* *ONA FSR: 2.279178 nm* *ng from ONA FSR: 4.221116*
S No 7 : *wgs_1 length: 60.000 um* *wgs_2 length: 310.000 um* *Delta L = abs(L2 - L1): 250.000 um* *Peak transmission: 1549.460188 nm, 1552.032662 nm* *Transmission FSR: 2.572474 nm at 1550.746425 nm* *ng from peak transmission: 3.739302* *ONA FSR sample nearest 1550 nm: 1549.460188 nm* *ONA FSR: 2.544040 nm* *ng from ONA FSR: 3.774825*	***S No 8 :*** *wgs_1 length: 147.133 um* *wgs_2 length: 492.736 um* *Delta L = abs(L2 - L1): 345.603 um* *ONA FSR sample nearest 1550 nm: 1550.36 nm* *ONA FSR: 0.829941 nm* *ng from ONA FSR: 4.19*
S No 9 : *wgs_1 length: 244.469 um* *wgs_2 length: 993.348 um* *Delta L = abs(L2 - L1): 748.879 um* *Peak transmission: 1549.880009 nm, 1550.310062 nm* *Transmission FSR: 0.430053 nm at 1550.095036 nm* *ng from peak transmission: 3.730382* *ONA FSR sample nearest 1550 nm: 1549.880009 nm* *ONA FSR: 0.429814 nm* *ng from ONA FSR: 3.7314165*	

XIV. EXPECTED DESIGN TREND

Effect of path-length difference

$$\boxed{\Delta L \uparrow \Rightarrow \text{FSR} \downarrow}$$

XV. FABRICATION DESIGN USING KLAYOUT

The circuits are subsequently laid out according to the design rules of the fabrication process.

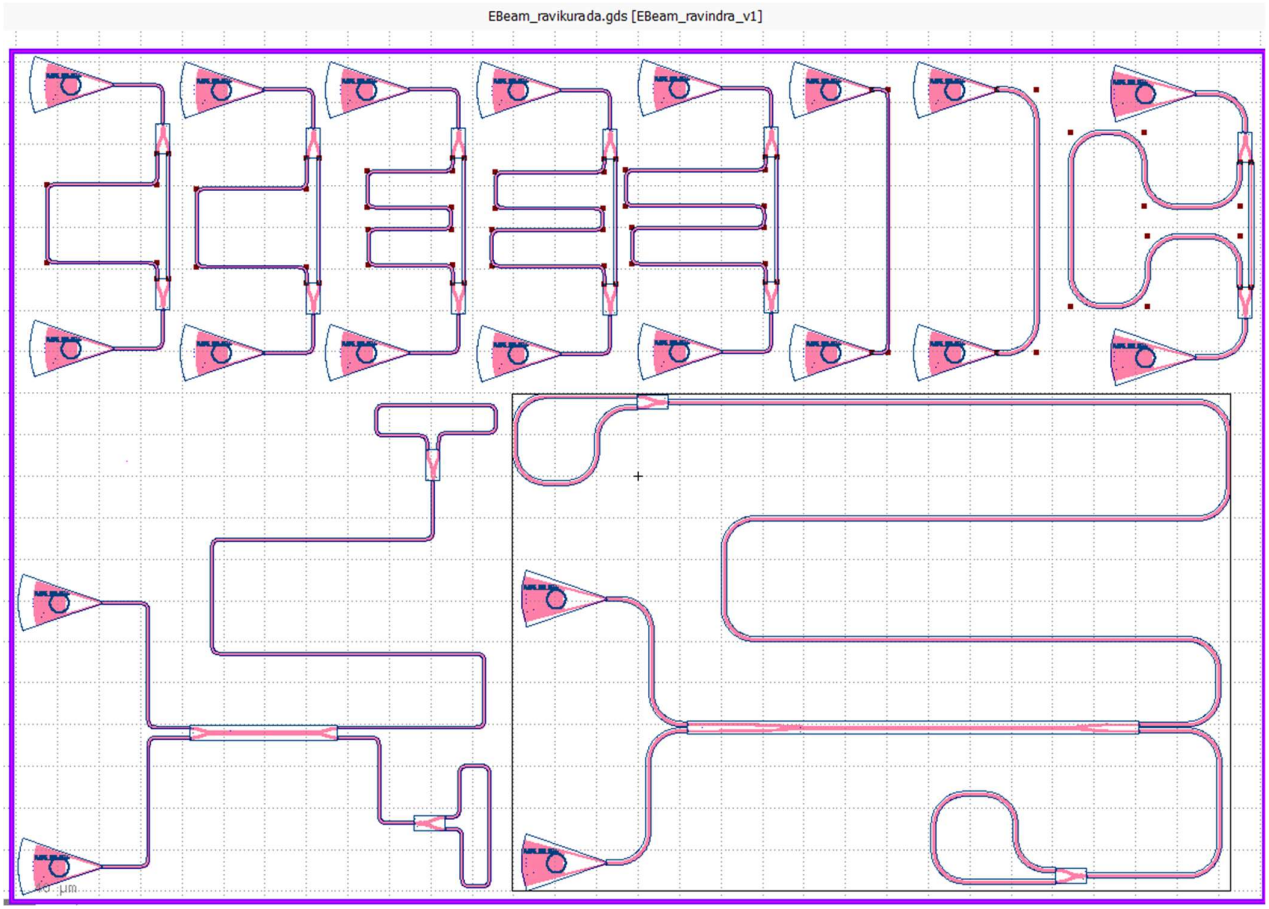


Fig. 13. Layout of the complete photonic test structures.

TABLE VI
SUMMARY OF SIMULATION RESULTS (KLAYOUT+LUMERICAL IC)

S No 1 & 2 : *wgs_1 length: 60.000 um* *wgs_2 length: 160.000 um* *Delta L = abs(L2 - L1): 100.000 um* *ONA FSR sample nearest 1550 nm: 1548.79 nm* *ONA FSR: 5.80974 nm* *ng from ONA FSR: 4.13*	***S No 3*** *:* *wgs_1 length: 60.000 um* *wgs_2 length: 210.000 um* *Delta L = abs(L2 - L1): 150.000 um* *ONA FSR sample nearest 1550 nm: 1548.75 nm* *ONA FSR: 3.88457 nm* *ng from ONA FSR: 4.12*
S No 4 : *wgs_1 length: 60.000 um* *wgs_2 length: 260.000 um* *Delta L = abs(L2 - L1): 200.000 um* *ONA FSR sample nearest 1550 nm: 1548.75 nm* *ONA FSR: 2.84923 nm* *ng from ONA FSR: 4.21*	***S No 5 :*** *wgs_1 length: 60.000 um* *wgs_2 length: 310.000 um* *Delta L = abs(L2 - L1): 250.000 um* *ONA FSR sample nearest 1550 nm: 1551.02 nm* *ONA FSR: 2.2869 nm* *ng from ONA FSR: 4.21*
S No 7 : *wgs_1 length: 60.000 um* *wgs_2 length: 310.000 um* *Delta L = abs(L2 - L1): 250.000 um* *ONA FSR sample nearest 1550 nm: 1550.43 nm* *ONA FSR: 2.55391 nm* *ng from ONA FSR: 3.76*	***S No 8 :*** *wgs_1 length: 75.188 um* *wgs_2 length: 420.791 um* *Delta L = abs(L2 - L1): 345.603 um* *ONA FSR sample nearest 1550 nm: 1550.4 nm* *ONA FSR: 0.83729 nm* *ng from ONA FSR: 4.15*
S No 9 : *wgs_1 length: 159.385 um* *wgs_2 length: 908.264 um* *Delta L = abs(L2 - L1): 748.879 um* *ONA FSR sample nearest 1550 nm: 1549.94 nm* *ONA FSR: 0.40569 nm* *ng from ONA FSR: 3.95*	

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